

जवाहरलाल नेहरू विश्वविद्यालय

रक्षा और अनुसंधान संस्थान अनुभाग

सं. डीआरआईएस/एस/पीएचडी/आईयूसीए/आईयूसीए/टीटी-14027/938-आर

सहायक कुलसचिव / उप कुलसचिव / संयुक्त कुलसचिव

दिनांक: 26-12-2024

सेवा में,

निदेशक

अंतर-विश्वविद्यालय केंद्र खगोलविज्ञान और खगोलभौतिकी, पुणे, (IUCAA)

नई दिल्ली

विषय: अंतर-विश्वविद्यालय केंद्र खगोलविज्ञान और खगोलभौतिकी, पुणे के श्री प्रेमविजय वी (पंजीकरण संख्या IUCAA/2019/001) द्वारा प्रस्तुत डॉक्टरल शोध-प्रबंध (Doctoral Thesis) की मूल्यांकन रिपोर्ट।

महोदय/महोदया,

कृपया उपर्युक्त छात्र द्वारा जमा की गई डॉक्टरल शोध-प्रबंध (Doctoral Thesis) पर परीक्षकों से प्राप्त रिपोर्ट और संस्तुतियां (मूल) आपके विचारार्थ और उचित कार्रवाई हेतु इसके साथ संलग्न कर भेजी जा रही हैं।

मौखिक परीक्षा बोर्ड के सदस्यों द्वारा उपयोग के लिए एक रिक्त फॉर्म भी संलग्न है।

अनुरोध है कि मौखिक परीक्षा बोर्ड, केंद्र की शैक्षिक समिति की सिफारिशों सहित रिपोर्ट कृपया अगली उचित कार्रवाई के लिए इस कार्यालय को यथाशीघ्र भेजी जाए।

सादर,

भवदीय

(सहायक कुलसचिव/उप कुलसचिव/ संयुक्त कुलसचिव)

संलग्नक: यथोपरि

(पृष्ठों की संख्या)

सुप्रिया
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JAWAHARLAL NEHRU UNIVERSITY
Defence and Research Institutions Section

Confidential

ASST. REGISTRAR / DY. REGISTRAR

No. DRIS / S / Ph.D. / IUCAA / IUCAA / TT - 14027 / 93

Date: 26-12-24

To,
The Director,
Inter-University Centre for Astronomy and Astrophysics, Pune

Subject: Evaluation report of the doctoral thesis submitted by Mr. Premvijay V (Enrol. No. IUCAA/2019/001) of Inter-University Centre for Astronomy and Astrophysics, Pune

Dear Sir / Madam,

Enclosed please find reports and recommendations (in original) received from examiners on the doctoral thesis submitted by the above-noted student for your consideration and appropriate action.

A blank form for use by members of the Viva-Voce Board is also enclosed.

It is requested that reports together with recommendations of the Viva-voce Board, Academic committee of the centre may please be forwarded to us as early as possible for further appropriate action.

With kind regards,

Yours faithfully

(ASST. REGISTRAR / DY. REGISTRAR)

Encl: As above
(No. of pages: ____)

E-mail: ri-incoming@jnu.ac.in / TEL.: 91-11-26704053



जवाहरलाल नेहरू विश्वविद्यालय
नई दिल्ली

JAWAHARLAL NEHRU UNIVERSITY

NEW DELHI - 110067

परीक्षक की सिफारिश

RECOMMENDATION OF EXAMINER

उम्मीदवार का नाम श्री / सुश्री
Name of the Candidate: Mr. / Ms. Premvijay V

लघु शोधप्रबंध का विषय
Title of the dissertation: Interplay of galaxy formation and the evolution of dark matter haioes
in the cosmic web

जिस उपाधि के लिए सिफारिश की जा रही है विद्या वाचस्पति (पी एच डी)
Degree for which recommendation is made: Doctor of Philosophy

मैं उक्त उम्मीदवार को मौखिक परीक्षा पास करने पर विद्या वाचस्पति (पी एच डी) की उपाधि प्रदान करने की
सिफारिश करता / करती हूँ।

I recommend the granting of the Degree of Doctor of Philosophy to the candidate subject to
his/her passing the Viva-Voce Examination.

हस्ताक्षर / Signature:

नाम / Name: Subhabrata Majumdar

पदनाम / Designation: Professor, DTP, TIFR, Mumbai

दिनांक / Date: 20 Dec 2024

Referee Report on thesis titled
**Interplay of galaxy formation and the evolution of dark matter haloes in the
cosmic web**

By PremVijay Velmani
submitted for award of PhD in Physics

A solid thesis, resulting out of top-quality publications, in the area of cosmology and dark matter halos. I consider this thesis one of the two best among all the theses that I have refereed in the past fifteen years. The work is of great value to contemporary galaxy/dark matter halos studies, both observations and simulations, and I **recommend acceptance of the thesis without any major modifications.**

The thesis focuses on spans aspects of cosmology and large-scale structure and galaxy-scale astrophysics using both cosmological hydrodynamical simulations (large particle simulations such as IllustrisTNG, EAGLE, and CAMELS) as well as numerical models of dark matter haloes with galaxies. For understanding, simulation results, Premvijay has present work using halo finding codes to generate halo catalogues with merger trees and and then used a KD-tree-based algorithm to match outputs from different simulations run with the same initial cosmological volumes. He has further used the particle data from these simulations for computing both large scale structure statistics as well as halo properties such as mass profiles and halo shapes.

The main aim is the investigation of the dynamics and evolution of dark matter haloes embedded in the cosmic web and their response to galaxy formation and evolution. The impact of the formation of galaxies on their host dark matter halos is inadequately studied in the literature, and Premvijay's thesis partially fill an important gap in this area of research. An enjoyable outcome of his effort are theoretical self-similar solutions to galaxy formation, interacting with its host dark matter halo. Overall, its an overall impressive work demonstrating complex numerical simulations skills, theoretical understanding, knowledge over analytical methods and promising progress made towards a synergy between halo formation/evolution and galaxy formation/evolution.

The research presented in this thesis is taken from three publications of Premvijay (two in JCAP and one in MNRAS), and thus has already undergone serious academic refereeing. He has another manuscript in preparation. Given the high quality of already published results, **I have only a few opinions/questions (listed below).**

1. One of the basic requirement for the work is to compare halos simulated in dark matter only simulations vs those having both dark matter and galaxies. Thus, for two halos in comparison, one has to define the "matching fraction of each of those two haloes (with respect to the other) as the fraction of its dark matter particles that are also present in the other halo". This is unclear to me. In the DM only simulations, this would mean all of the halo, whereas for hydro simulations, this is just the dark matter part of the halo. In such case, do the "total mass" (defined as, say, M_{200}) of the halo remain the same? How does one divide between baryonic mass/dark matter mass/total mass of simulated halos to compare in a correct apple-to-apple manner with theoretical descriptions, and even more so with observations?

Also, how does it affect results when using merger trees, halo catalogues and

Srinivasan

statistical description of halos.

The answer might be obvious which I have missed, but explaining this during the viva and clarifying it in the text would be helpful.

2. The thesis deals mainly with the density/mass profiles and shapes of halos, and also addresses aspects of velocity profiles. However, given the impact that galaxy formation has on halo properties, it has little on the impact of dark matter velocity distribution function which is a key in all astro-particle probes of dark matter in halos. I also did not find any discussion of the dark matter phase-space (or pseudo phase-space) which is crucial output of halo simulations as well as theoretical models like secondary infall model.
3. In general, there is little comparison with simulations, throughout the thesis. I think that is a pity. For example, there is only a sub-sub-section on rotation curves (and none on velocity dispersion via Jeans equation). Given that rotation curves are the crucial to connecting observed halos with theory and serve as the central step connecting observed dark matter phase-space to theory/simulations, this is one major part missing from an otherwise remarkable thesis.

Apart from these are a number of minor questions which can be asked during the viva. I have also not commented on grammar and typos. In conclusion, this thesis was a pleasure to read and examine, and I recommend its acceptance without any hesitations.



20th December 2024

Prof Subhabrata Majumdar
Professor
DTP, TIFR, Colaba
Mumbai 400005
email: subha@tifr.res.in
subha@theory.tifr.res.in



tokgjYkkYk usg: fo'ofokYk;

नई द ली 110067

JAWAHARLAL NEHRU UNIVERSITY

NEW DELHI - 110067

RECOMMENDATION OF EXAMINER

Name of the Candidate:

Mr./Ms. ...Premvijay Velmani

Title of the Thesis:

...Interplay of galaxy formation and the evolution
of dark matter haloes in the cosmic web.....

Degree for which the
Recommendation is made:

DOCTOR OF PHILOSOPHY

- i) *I recommend the granting of the Degree of Doctor of Philosophy to
the candidate subject to his/her passing the Viva-Voce Examination.

Signature:

Name: Raul E. Angulo

Designation:

Date

*Please strike off whichever is not applicable.



Donostia-San Sebastian, November 2024

Report for the thesis of Premvijay Velmani

I have carefully read the thesis "*Interplay of galaxy formation and the evolution of dark matter haloes in the cosmic web*" submitted by Mr. Premvijay Velmani in partial fulfilment of the requirements for the degree of PhD in Physics.

The main topic of the thesis is studying the role of astrophysical processes in shaping the underlying distribution of dark matter. To achieve this, Mr Velmani analysed multiple hydrodynamical simulations, proposed an extension of the adiabatic contraction model, and developed an analytical model to understand how star formation and cooling would affect the density profile of dark matter inside collapsed haloes. Quantifying how astrophysics modifies the distribution of dark matter is essential in modern cosmology since detailed modelling is crucial for interpreting observations aimed at distinguishing different dark matter candidates and cosmological models.

The thesis is structured in 7 chapters. The first two chapters provide an introduction to the thesis and summarise the simulations and methodologies employed. Specifically, these chapters describe the various hydrodynamical simulations analysed, the matching between full-physics and gravity-only runs, the measurement of the dark matter response to baryonic processes, and provide a recap of the state of the art and the main ideas behind theoretical models.

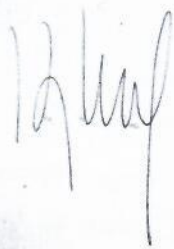
The following three chapters measure the response of dark matter to galaxy formation processes in different simulations and redshifts. First, the dark matter response was measured in the IllustrisTNG simulation and compared with the expectations of relaxation models found in the literature. Since these models failed to capture the simulation results accurately, an extension was proposed that, with additional free parameters, was able to describe the IllustrisTNG's results in multiple halo masses, and also in other hydrodynamical simulations such

as EAGLE and CAMELS, as well as at higher redshifts. Through a time-analysis of dark matter relaxation across time, the additional free parameters of the model were interpreted as quantifying astrophysical processes such as gas ejection and star formation, which could modify the relaxation process and the amount of mass at a given location.

Chapter 6 presents a theoretical model for these effects. Specifically, the chapter provides a novel extension of a self-similar model that includes gas cooling and star formation to understand the results provided in previous chapters qualitatively. The final chapter summarises the main findings of this thesis and provides an outlook for possible applications in other areas of cosmology.

Overall, the thesis is a novel and an essential contribution to the field of cosmology, and it represents an important step towards building reliable predictions for our cosmological models. This is central in modern cosmology, where interpreting observations such as weak gravitational lensing and rotation curves in galaxies will require accurate theoretical modelling. Additionally, the careful study of halo relaxation in hydrodynamical simulations will raise awareness of the limitations of current models and the importance of improving their accuracy as well as accounting for their uncertainty. Finally, I expect the results to impact the field and motivate further research. For instance, the proposed extension to relaxation processes can be incorporated into baryonification algorithms to improve their reliability and marginalise over these uncertainties, leading to more robust cosmological constraints. Ultimately, such models could place constraints on relaxation processes, thus informing about the assembly of stars and gas in the universe.

Therefore, in my opinion, this manuscript fulfils all the requirements of depth and originality of a PhD thesis, and **I recommend the granting of the Degree of Doctor of Philosophy to the candidate subject to his passing the Viva-Voce Examination.**



Raul Angulo
Research Associate
Donostia International Physics Center